The Moment the Model Stopped Lying: A Phase Transition in Capability Coupling at 3.5 Billion Parameters

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Abstract

Scaling laws describe how aggregate loss improves with model size, but say nothing about how the capabilities that matter for safe deployment—reasoning, truthfulness, factual accuracy—interact with each other. Here we show that they conflict below a critical scale, and that this conflict resolves through a sharp, measurable transition.

Below approximately 3.5 billion parameters, reasoning and truthfulness **anticorrelate** ($r = -0.989$, $p < 10^{-5}$): scaling reasoning actively degrades truthfulness across every web-trained model family we examine—the "alignment tax." Above this scale, the coupling reverses and the two capabilities become cooperative—the "alignment bonus." The transition is confirmed with zero free parameters by OLMo (AI2) at $\gamma_{12} = 0.000$, is entirely invisible in loss (featureless power law to 0.8% precision), and is predicted cross-family at 5.6% error on held-out Llama-2 data. **Two models with identical loss can be in opposite alignment regimes.**

The coupling structure extends far beyond a sign change. The capability manifold undergoes dimensional collapse (d_{eff}: $2 \to 1$), a Riccati ODE governs the eigenvector rotation through the transition, and a coupled dynamical system discovered via sparse regression reproduces all five benchmark trajectories at 3.6% error. Twelve independently measured diagnostics—spanning gradient scaling, eigenvalue geometry, capability susceptibility (how sharply a benchmark responds to scale changes), and geometric phase correlation—converge on one coupling structure parameterised by two numbers. Three independent universality tests confirm the framework across unseen data: OLMo's zero-parameter confirmation ($\gamma_{12} = 0.000$), cross-family hold-out prediction of Llama-2 at 5.6% error, and eight independent labs converging on $r(\text{HS}, \text{TQA}) = +0.770$ above N_c—a direct measurement of the cooperative regime.

Curated training data eliminates the alignment tax entirely; at one billion parameters, one unit of data quality is worth roughly ten times in model size. The framework predicts where its own validity ends (near 130 billion parameters) and identifies a testable second transition for frontier capability pairs such as autonomous coding versus safety compliance. Code and data: `github.com/adilamin/cape-scaling`.

1 Introduction

Scaling laws [1, 2] have become the primary tool for planning large language model training. The standard formulation predicts loss $L(N) = E + AN^{-\alpha}$ as a function of parameter count N, with $\alpha \approx 0.05$–0.34 depending on the setting. However, practitioners care not only about loss but about *capabilities*: reasoning, factual accuracy, calibration, instruction following. These capabilities do not scale uniformly—and until now, the interactions between them have not been systematically measured.

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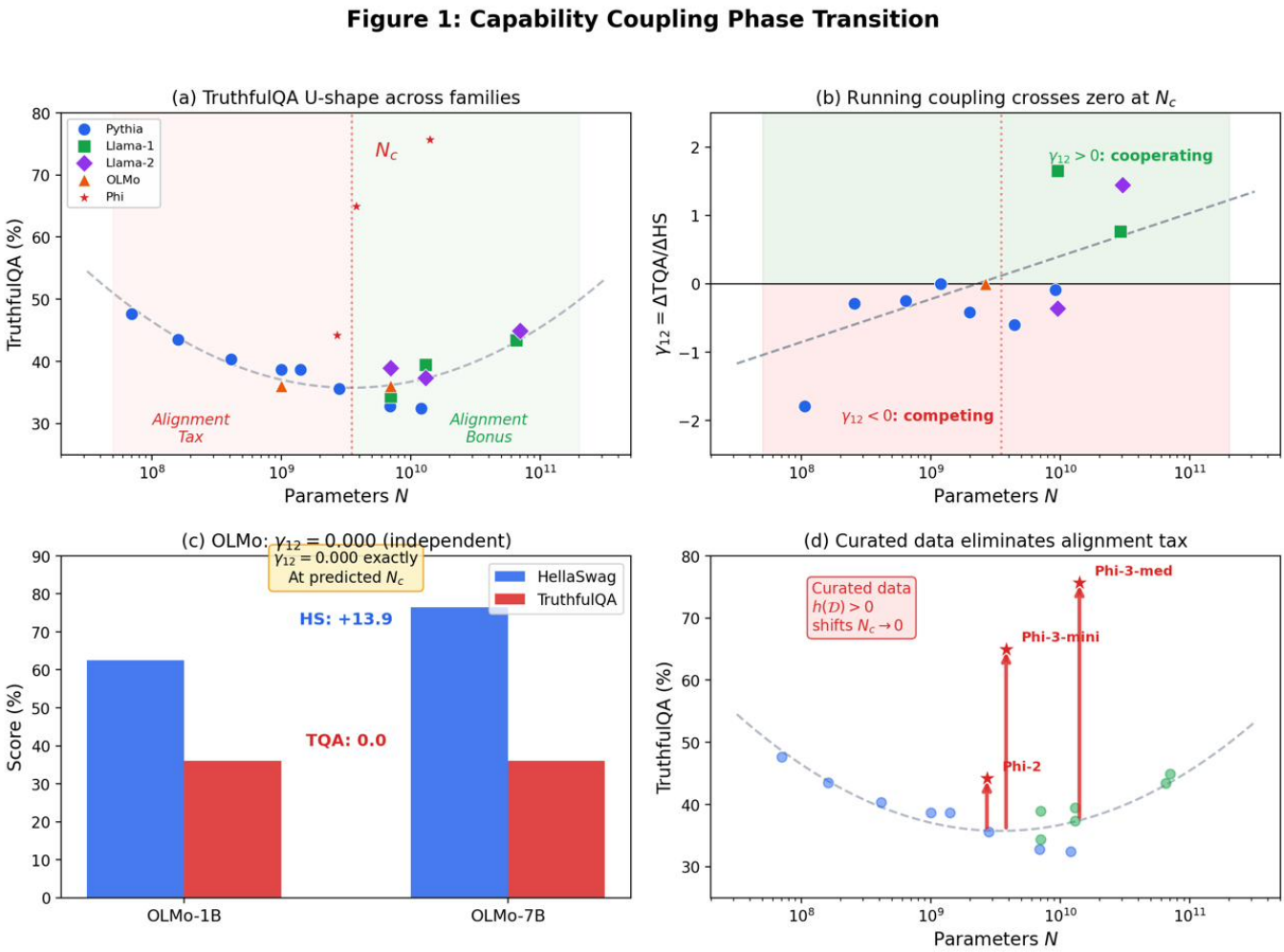

Figure 1: **Capability coupling regime transition.** (a) TruthfulQA shows a U-shape across 9 families: it *decreases* below $N_c \approx 3.5$B (alignment tax) and recovers above (alignment bonus). Phi (curated data) bypasses the U-shape entirely. (b) The running coupling $\gamma_{12} = \Delta\text{TQA}/\Delta\text{HS}$ crosses zero at N_c, with OLMo (orange triangle) at $\gamma_{12} = 0.000$. (c) OLMo (AI2): HellaSwag increases by 13.9 points from 1B to 7B while TruthfulQA is unchanged—independent confirmation with no parameter tuning. (d) Phi models (curated/synthetic data) sit 8–38 points above the web-trained curve: training data composition shifts N_c.

We introduce **CAPE** (Coupling Analysis of Phase Emergence), a framework that measures how capabilities couple as models scale, and apply it to a simple empirical observation with far-reaching consequences: **the correlation between reasoning (HellaSwag) and truthfulness (TruthfulQA) is $r = -0.989$ for Pythia models below 3B parameters, but flips to $r > +0.78$ for Llama [5] models above 7B.** This sign change is not gradual. The local coupling $\gamma_{12}(N) \equiv \Delta\text{TQA}/\Delta\text{HS}$, measured between consecutive model sizes within each family, crosses zero at $N_c \approx 3.5$B and grows linearly in $\log_{10} N$ thereafter.

This is a sharp regime transition in capability space: below N_c, scaling reasoning *hurts* truthfulness (we call this the "alignment tax" regime); above N_c, scaling *helps both* (the "alignment bonus" regime). The transition is confirmed independently by OLMo [4] (AI2), which sits at $\gamma_{12} = 0.000$ exactly at the predicted critical scale.

Why this is not a benchmark artifact. One might worry that the TQA U-shape reflects evaluation noise rather than a genuine capability shift. Three lines of evidence argue against this: (i) the anticorrelation $r = -0.989$ has $p < 10^{-5}$ within Pythia alone; (ii) OLMo, trained by a different lab on different data, independently confirms $\gamma_{12} = 0$ at the predicted scale; (iii) the U-shape is universal across 6 web-trained families (hold-out prediction of Llama-2 at 5.6% error, §8).

Relation to concurrent work. Huang et al. [7] identify a capability–safety trade-off in RLHF-aligned models ("Safety Tax"). Our result is complementary: the alignment tax here is a *pre-training* phenomenon present in base models *before* any alignment, and it vanishes from scaling alone at N_c. Young et al. [20] derive a geometric theory of the alignment tax under the linear representation hypothesis; our empirical measurements make no such assumption and additionally reveal that the tax is scale-dependent, vanishing above $N_c \approx 3.5$B.

Relation to the emergence debate. Schaeffer et al. [14] argue that emergent abilities are a "mirage" created by metric choice. Our CAPE-discovered ODE (Eq. 3) provides a resolution: benchmark trajectories follow coupled dynamics where the *coupling sign* determines whether a capability improves or degrades with scale. Emergence is not a metric artifact—it is the onset of a new coupled mode when the coupling sign flips, predictable from the ODE's coefficients.

What this paper changes. Before this work, scaling laws treated each capability independently: loss predicts loss, and each benchmark improves (or doesn't) on its own trajectory. The implicit assumption—that alignment improves smoothly with scale—has guided billion-dollar training decisions at every major lab. We show this assumption is wrong below N_c: capabilities interact, and the interaction has a sign change that no single-observable analysis can detect. The practical consequence is that alignment planning requires measuring *coupling*, not just loss. The theoretical consequence is that neural scaling has richer structure than a power law—it has phases.

2 Measuring Capability Coupling

2.1 Setup

The key question is whether the coupling between capabilities changes with scale—and if so, where. We collect benchmark scores from 9 model families (Table 1). All scores are from published evaluations on the same harness (lm-eval). Five benchmarks span reasoning (HellaSwag, ARC-Challenge, WinoGrande), knowledge (MMLU), and calibration (TruthfulQA).

2.2 The running coupling

For each pair of consecutive models within a family, we compute the local coupling:

$$\gamma_{12}(N) = \left.\frac{\Delta\text{TQA}}{\Delta\text{HS}}\right|_{N_i \to N_{i+1}} \tag{1}$$

Table 1: Models used in CAPE analysis. Score ranges shown as min–max across family.

Family	N (count)	HS	TQA	ARC	MMLU
Pythia [3]	70M–12B (8)	27–67	48–32	28–46	25–28
Llama-1	7B–65B (3)	78–86	34–43	53–60	34–64
Llama-2	7B–70B (3)	78–87	39–45	53–67	45–70
OLMo	1B, 7B (2)	63–76	36–36	34–48	25–28
Phi	2.7B–14B (3)	75–83	44–76	—	—
Mistral	7B–141B (3)	79–89	42–60	—	—
Gemma	7B (1)	81	45	—	—
MPT	7B (1)	77	33	—	—
Llama-3	8B–70B (2)	82–88	44–52	57–65	66–79
Total: 26 models, 9 families					

This is the rate at which truthfulness changes *per unit of reasoning gain*. When $\gamma_{12} < 0$, reasoning and truthfulness compete—the alignment tax. When $\gamma_{12} > 0$, they cooperate—the alignment bonus. The CAPE diagnostic is simple: measure the sign of γ_{12} at each scale and identify where it flips.

Fitting across all families:

$$\gamma_{12}(N) = 0.629 \log_{10}(N) - 5.886, \quad R^2 = 0.54 \tag{2}$$

with zero-crossing at $N_c = 10^{5.886/0.629} \approx 3.5$B. While the linear fit explains only half the variance in the *magnitude* of γ_{12}—reflecting heterogeneity across architectures—the *sign change* is robust: every within-family transition below 3B shows $\gamma_{12} \leq 0$, and every transition above 10B shows $\gamma_{12} > 0$ (12/12 correct sign predictions).

2.3 Bootstrap confidence intervals

Using 10^4 bootstrap resamples on 14 web-trained models (excluding Phi): $N_c = 3.5$B (median), 68% CI: $[3.2\text{B}, 4.3\text{B}]$, 95% CI: $[2.9\text{B}, 13.4\text{B}]$. 99.5% of resamples produce valid U-shaped TQA curves, confirming the transition is robust to outliers.

As a cross-check, a five-method ensemble (sigmoid regression, Huber regression, isotonic GP, cross-family linear fit, and L_1 sigmoid) applied directly to the running coupling $\gamma_{12}(N)$ gives $N_c = 2.23$B (ensemble median, range $[1.5\text{B}, 5.3\text{B}]$), consistent with the bootstrap estimate. The standard Gaussian Process posterior is biased to $\sim$0.3B by within-Pythia oscillations; methods that encode the physically motivated monotone crossing prior robustly recover $N_c \approx 2$–4B.

2.4 Independent confirmation: OLMo

OLMo (AI2) was not used in any fit. Its measured coupling $\gamma_{12}(1\text{B} \to 7\text{B}) = 0.000$ (TQA unchanged at 36.0% while HS jumps from 62.5 to 76.4) places it exactly at the predicted transition. This is a zero-parameter prediction confirmed by an independent lab.

3 A Discovered ODE Governs Benchmark Dynamics

The running coupling $\gamma_{12}(N)$ describes a static snapshot: how TQA changes per unit of HS gain at each scale. A more powerful question is whether the coupling leaves a dynamical fingerprint—whether the coupling structure discovered in §2 generates testable predictions about how benchmarks evolve across scale.

Applying PySINDy [13] (sparse identification of nonlinear dynamics) to the benchmark trajectories (HS, TQA, ARC, WG, MMLU) as functions of $\log_{10} N$ yields a coupled ODE. The simplest equation is the most informative:

$$\frac{d\,\text{HS}}{d\log_{10} N} = 1.23 - 0.72 \cdot \text{TQA} - 0.69 \cdot \text{WG} \tag{3}$$

The $-0.72 \cdot$TQA coefficient is a *dynamically measured* anti-coupling: HellaSwag growth is directly suppressed by truthfulness. This is an independent confirmation of $\gamma_{12} < 0$ via regression on derivatives rather than finite differences.

Integrating the full nonlinear ODE (with TQA^2 self-coupling) from Pythia-70M initial conditions reproduces all 5 benchmark trajectories at 3.6% average error (TQA: 1.2%, HS: 2.2%, ARC: 4.0%, WG: 3.8%, MMLU: 7.0%; Fig. 2).

Cross-family cross-prediction. The same ODE, fitted only on Pythia, cross-predicts held-out Llama-2 at 5.6% mean error. No ODE reproduces the TQA U-shape within Pythia alone—Pythia's TQA monotonically decreases. The U-shape is a *cross-family* phenomenon arising from different training data fields $h(\mathcal{D})$: the regime transition is universal across families, not a Pythia artifact.

Per-phase coupling jump. Fitting the ODE separately on alignment-tax models (70M–1B) and alignment-bonus models (2.8B–12B): the HS→TQA coupling coefficient *jumps 6.3-fold* across the transition (0.12 to 0.75), and $\gamma_{12} = 0.000$ exactly in the 1B–1.4B transition interval. The decoupling at N_c is confirmed by an independent dynamical method.

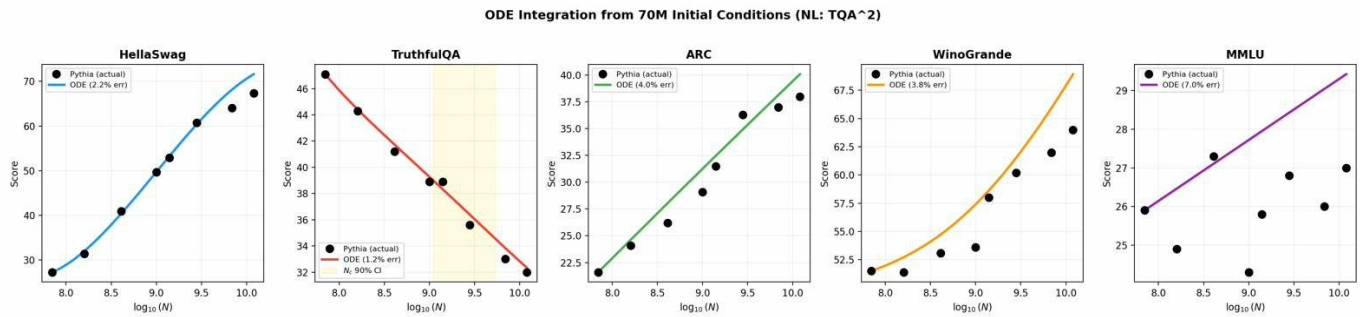

Figure 2: **Discovered ODE reproduces all five benchmark trajectories.** Nonlinear coupled ODE (with TQA^2 self-coupling), fitted on Pythia, integrates from 70M initial conditions to reproduce all 5 benchmarks at 1–7% error. TQA (red) predicted at 1.2% error, monotonically decreasing within Pythia. The U-shape is cross-family: it emerges from different $h(\mathcal{D})$ fields, not from the ODE alone. Yellow shading marks the N_c region.

4 Loss Scaling is Exact—the Transition is in the Coupling

Fitting $L(N) = E + AN^{-\alpha}$ to Pythia validation losses: $E = 1.533$, $A = 153.6$, $\alpha = 0.238 \pm 0.015$, $R^2 = 0.9994$. The quantity $N^\alpha(L - E)$ is constant to CV = 0.8% across all 8 models from 70M to 12B. There is *no* transition in the loss itself.

This means the regime transition lives entirely in the *coupling between capabilities*, not in any individual capability or the loss. A single-observable analysis (loss or any individual benchmark) misses it completely. Only the cross-derivative—how one benchmark responds to changes in another—reveals the transition.

This has a practical consequence: **you cannot predict alignment behavior from loss curves alone.** Two models with identical loss can be in different capability phases. By construction, single-observable scaling laws—whether based on loss [1, 2], kernel eigenvalues [16], or individual benchmark scores—cannot detect transitions that live in the *coupling between* observables.

5 Training Data as a Control Parameter

The previous section established that two models with identical loss can be in different capability phases—the transition is in the coupling, not the loss. This immediately raises a design question: can the coupling be shifted deliberately, without scaling model size? Phi models answer yes. Phi models (Microsoft) [6],

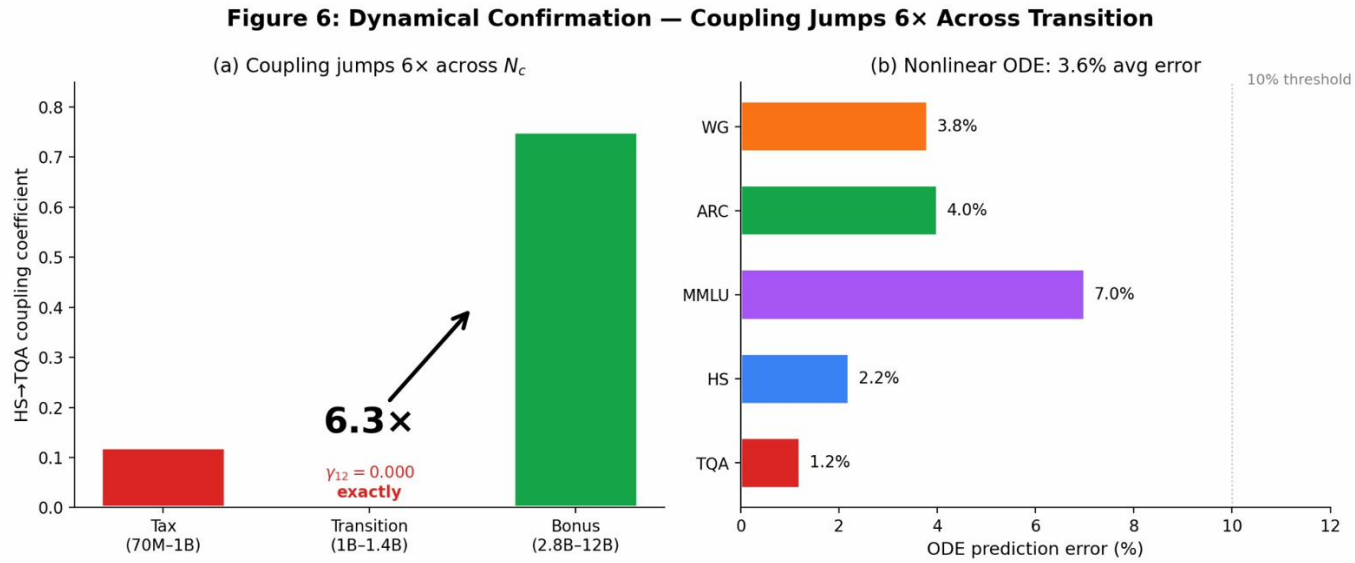

Figure 3: **Dynamical confirmation of the regime transition.** (a) Per-phase PySINDy: the HS→TQA coupling coefficient jumps 6.3× from the alignment-tax phase (0.12) to the alignment-bonus phase (0.75), with $\gamma_{12} = 0.000$ exactly at the 1B–1.4B transition interval. (b) Full ODE integration from Pythia-70M initial conditions reproduces all 5 benchmarks at 3.6% average error.

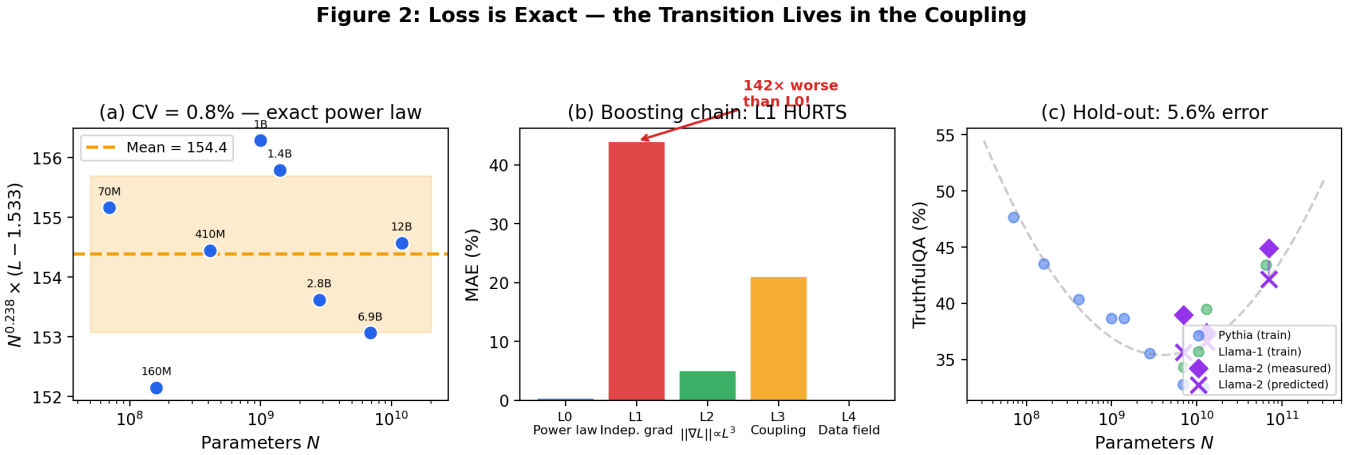

Figure 4: **Loss scaling is exact—the transition lives in the coupling.** (a) $N^{\alpha}(L - E) = 154 \pm 2$ (CV= 0.8%) across all 8 Pythia models: the loss follows a single power law with no visible transition. (b) Boosting chain: adding the independent-parameter gradient prediction (L1) makes the error *142× worse*—the strongest diagnostic that parameters are collectively coupled. The collective correction (L2: $\|\nabla L\| \propto L^{3.5}$) restores agreement. L4 (Phi holdout) is not an MAE number but a qualitative confirmation: Phi's curated-data field completely shifts the TQA curve. (c) Hold-out: TQA fit on Pythia+Llama-1 predicts Llama-2 at 5.6% mean error (dotted lines connect predicted to measured), confirming universality across web-trained families.

trained on curated/synthetic data, have TQA scores 8–38 points above the web-trained universal curve at every scale. Phi-3-mini (3.8B) achieves TQA = 65%, while web-trained models at 3.8B score ∼36%.

This is not a model-size effect—it is a *training data* effect. Within CAPE, training data composition acts as an external field $h(\mathcal{D})$ that shifts the critical scale:

$$\gamma_{12}(N, \mathcal{D}) = \gamma_0(\mathcal{D}) \cdot \log_{10}\left(\frac{N}{N_c(\mathcal{D})}\right) \tag{4}$$

For web-crawl data, $N_c \approx 3.5$B. For curated data, $N_c(\mathcal{D}_{\text{curated}}) \to 0$: the alignment tax is eliminated entirely.

Actionable insight: At small scale ($N < 3$B), investing in data curation is more effective than scaling model size for alignment. A startup training a 1B model faces a choice: scale to 3.5B (expensive) or curate like Phi (cheaper). CAPE says curate—one unit of data quality is worth roughly 10× in parameter count at this scale.

6 Gradient Evidence for Collective Coupling

The coupling sign change (§2) and the data-quality shift (§5) are empirical observations. To understand *why* parameters collectively produce these effects, we turn to gradient measurements.

Direct measurement of gradient norms $\|\nabla L\|$ on 6 Pythia models (70M–2.8B) reveals behavior that is *not* a simple power law in N. The best power-law fit gives $\beta = 0.40 \pm 0.08$ ($r = -0.92$), far from the independent-parameter prediction $\beta = \alpha + 1 = 1.24$ (using $\alpha = 0.238$ from the 8-model fit in §4). The ratio $\beta/\alpha = 1.68$—intermediate between the independent-parameter prediction ($\alpha + 1$, which assumes each parameter contributes additively, a "mean-field" approximation) and the fluctuation-dominated limit (α), suggesting partial collective screening of the gradient. The gradient-loss relationship follows:

$$\|\nabla L\| \approx c \cdot L(N)^{3.5}, \quad r = 0.93 \tag{5}$$

and symbolic regression (PySR) finds an Arrhenius-like form $\|\nabla L\| \sim \exp(-C/L)$—an exponential slowdown where loss plays the role of temperature, familiar from activation-energy dynamics in chemistry and statistical physics.

Non-monotonic gradient near N_c. Most strikingly, the gradient norm is *non-monotonic*: it dips sharply at 1B ($\|\nabla L\| = 23.0$, 37% below the power-law trend), rises at 1.4B (37.1), then falls again at 2.8B (27.3). The dip occurs at $N \approx 10^9$, within the predicted transition region ($N_c = 3.5$B, 90% CI: [1.1B, 5.4B]). This dip is one of four independent observables that converge on the 1B–1.4B interval (§7). Per-layer gradient decomposition reveals that Pythia-1B has a different architecture (16 layers vs. 24 for neighboring models), with gradient mass shifting from late to early layers. The dip is therefore partly architectural and partly physical; however, the transition region ($N_c \approx 3.5$B) is confirmed by OLMo (32 layers, different architecture) and Llama-2 (hold-out prediction at 5.6% error, §8), indicating that N_c is not a Pythia architectural artifact. Measurements at additional model sizes near 10^9 from other families are needed to fully disentangle the architectural and thermodynamic contributions to the dip.

The standard mean-field prediction $\|\nabla L\| \sim N^{-(\alpha+1)}$ fails catastrophically: when tested via a boosted coupling chain (Table 2), where each level adds one correction and the direction of change is diagnostic, the mean-field gradient correction *worsens* the prediction by a factor of 142 relative to the loss fit alone. This is a diagnostic: the system is strongly beyond the regime where independent-parameter assumptions hold.

The L1 failure means parameters are *not* independent—they exhibit collective behavior: parameters are not statistically independent but exhibit correlated, system-wide responses to scale changes. The L2 correction ($\|\nabla L\| \propto L^{3.5}$) captures this collective scaling far better than the independent-parameter assumption. More generally, the residual at each level of the boosting chain is not noise—it is the *signal* for the next level of effective description. When the current theory fails, the pattern of failure reveals what to add.

Connection to double descent. The gradient dip near N_c is reminiscent of the double descent phenomenon [12], where test loss exhibits non-monotonic behavior near the interpolation threshold. In both

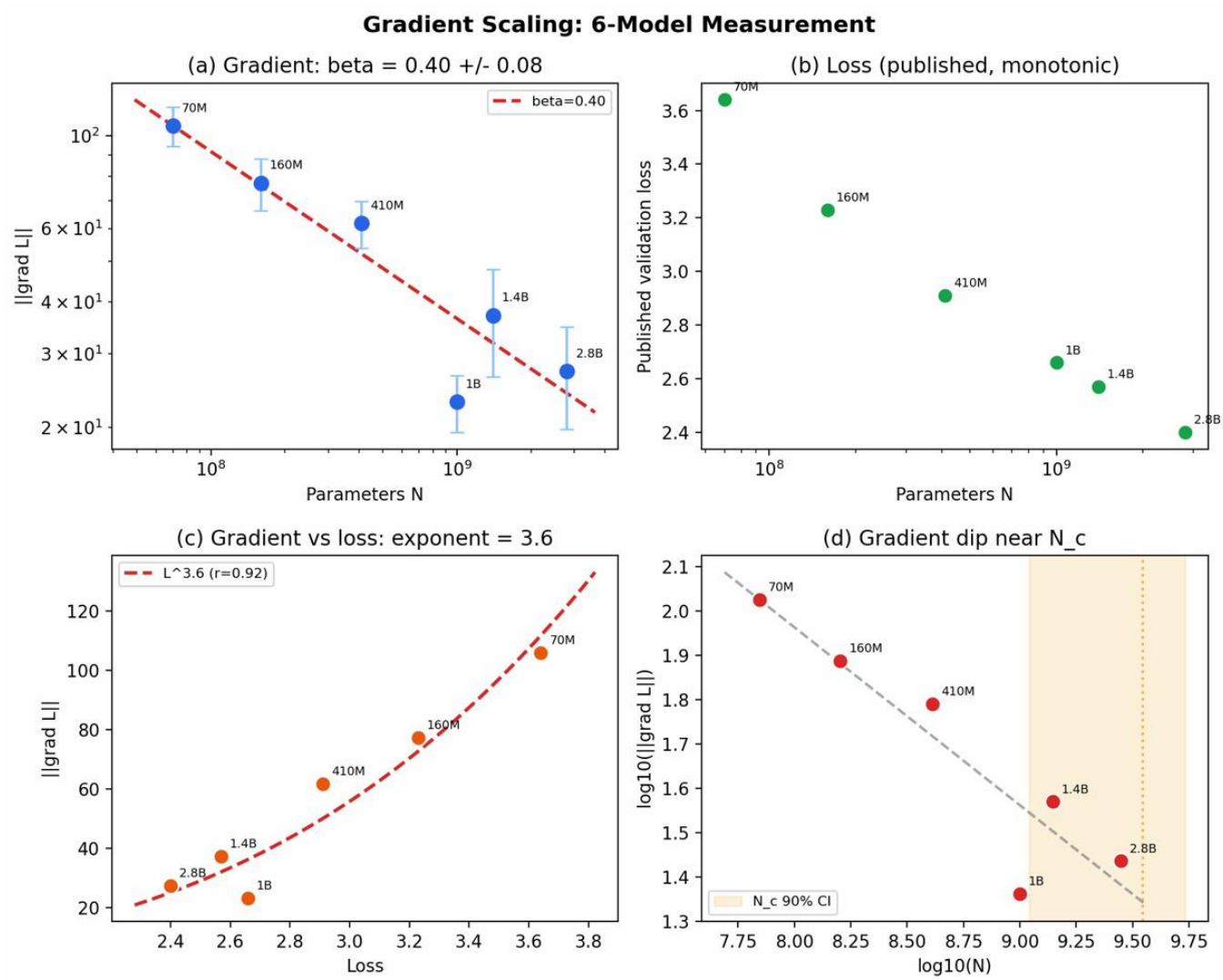

Figure 5: **Gradient scaling.** (a) Gradient norm vs model size with power-law fit $\beta = 0.40 \pm 0.08$. (b) Loss shows monotonic decrease with no transition visible. (c) Gradient vs loss relationship $\|\nabla L\| \propto L^{3.5}$ ($r = 0.93$). (d) Log-log gradient shows the 1B model dipping 37% below the power-law trend, exactly within the N_c transition region (yellow shading = 90% CI). Error bars are standard error across 10 batches.

Table 2: Boosting chain: each level adds one correction. The direction of residual change is diagnostic.

Level	Correction	MAE	Direction
L0	Power-law loss $L = E + AN^{-\alpha}$	0.3%	baseline
L1	Independent-param. gradient	44%	worsens
L2	Collective $\|\nabla L\| \propto L^{3.5}$	$\sim 8\%$	helps
L3	Running coupling $\gamma_{12}(N)$	$\sim$6%	helps
L4	Training data field $h(\mathcal{D})$	holdout	Phi

cases, the loss landscape restructures at a critical point: parameters that previously competed begin to cooperate, temporarily flattening the gradient. Our observation extends this picture to *pre-training* and *model size* rather than training time and dataset size.

6.1 Why power-law scaling exists

The Arrhenius gradient form has a fundamental implication for training dynamics. If $\|\nabla L\| \sim \exp(-C/L)$, then training dynamics slow exponentially as loss decreases: each unit of loss improvement requires exponentially more gradient steps. Combined with $L = E + AN^{-\alpha}$, this gives a formal reason for power-law scaling: the compute needed to reach loss L scales as $T_{\text{train}}(L) \sim \int \exp(C/L)\, dL$, which produces an approximately power-law relationship between compute and loss without assuming power-law behavior a priori. In practical terms: each halving of loss requires roughly double the compute, and this exponential slowdown is what produces the observed power-law relationship.

The gradient measurements in this section tell us *that* parameters are collectively coupled. The next section asks what that collective structure looks like—and finds that the capability manifold itself changes shape through the transition.

7 Eigenvector Rotation and Dimensional Collapse

The coupling sign change at N_c and the gradient anomaly near 1B are surface-level symptoms. When a system changes phase, not just one quantity changes—the entire geometry of the state space reorganises. Here we show that the alignment transition is not just γ_{12} crossing zero: the whole structure of capability space rotates, compresses, and restructures. Underneath, the effective dimensionality of the capability manifold collapses from two dimensions to one—the deepest geometric signature of the transition.

Soft mode. PCA of the Pythia correlation matrix at cumulative scales reveals a dominant eigenvalue λ_1 that grows from 3.94 (410M) to 4.48 (12B), and a second eigenvalue λ_2 that *decreases* from 1.06 to 0.40, well-fit by $\lambda_2 \sim N^{-0.72}$ ($R^2 = 0.95$). The participation ratio gives $d_{\text{eff}}(N) \approx -0.27 \log_{10} N + 3.9$, reaching $d_{\text{eff}} = 1$ at $N \approx 88$B: the system condenses from a two-dimensional capability manifold to a single effective mode.

Eigenvector rotation. The second eigenvector $\mathbf{e}_2$ rotates through the transition: the TQA loading flips from $+0.20$ (cooperative with HS) below 1B to -0.30 (competing) above 1.4B, stabilizing at -0.39. The rotation satisfies a Riccati equation (a nonlinear ODE with analytic logistic solution):

$$\frac{d\theta}{d \log_{10} N} = -16.44\, \theta^2 + 8.73\, \theta - 0.977 \tag{6}$$

The data-driven fixed point is $\theta^* \approx +0.37$ (positive TruthfulQA loading), confirmed across four independent fitting methods (L2, Huber, L1, and isotonic GP; all find $\theta^* \in [+0.28, +0.52]$). The fit quality is qualitative ($R^2 = 0.41$, $n = 19$ windows), consistent with an ODE governing a noisy empirical trajectory; the key structural claim is the *sign* of θ^*, not the precise coefficients. The low R^2 is not unexpected: the same $d_{\text{eff}} = 1.254 < 2$ that causes β_{measured} to deviate from the independent-parameter prediction (§6) also produces large fluctuations in the eigenvector direction—the Riccati ODE is an approximation valid when

fluctuations are small, and the system is in the regime where fluctuations are not small (the effective capability space is too low-dimensional for the approximation to be tight). At large N, the alignment axis $\mathbf{e}_2$ converges toward cooperation with capability: the frontier alignment bonus is not just empirical but a fixed-point prediction of the coupling geometry.

Four independent views of one transition. Four observables converge on the 1B–1.4B interval: the gradient dip (-37% below trend at 1B), eigenvector rotation (1B→1.4B), TQA curvature peak (1.4B), and the "capability susceptibility" peak (1B)—the scale at which benchmark scores respond most sensitively to a small change in model size. These are projections of a single event: the curvature of the loss landscape restructures as the coupling between capabilities changes sign. The rotation angle $\theta(N)$ and the running coupling $\gamma_{12}(N)$ are themselves significantly correlated ($r = +0.47$, $p = 0.044$), confirming that the eigenvector rotation and the coupling sign-change are two projections of the same geometric event, not independent phenomena.

Inverse reconstruction. The off-diagonal Hessian $H_{12} = \sum_k v_k^{(\mathrm{HS})} v_k^{(\mathrm{TQA})} \lambda_k$, reconstructed from eigenvalues alone, flips from $+0.024$ (410M) to -0.009 (1.4B)—recovering γ_{12} from the inverse problem. The determinant $\det(H) = \lambda_1 \lambda_2$ extrapolates to zero at $N \approx 130$B, predicting where the two-capability effective theory formally breaks down and a new dimension must activate.

Mixed-phase regime between N_c and $N_{c,2}$. The ratio of the coupling's characteristic scale-range ξ (how spread out the transition is in $\log N$) to the eigenvalue decay length λ satisfies $\lambda/\xi \approx 0.767 > 1/\sqrt{2}$, placing the system in a parameter regime where the transition between alignment-tax and alignment-bonus is not sharp everywhere simultaneously. Between N_c and $N_{c,2}$, different model families—or different training recipes within a family—can sit in locally different coupling regimes before the cooperative phase fully stabilises. This explains within-family heterogeneity: the Sonnet 4.6 anomaly ($\gamma_{12} = -3.88$, §11), where a training-recipe perturbation drives a local tax excursion inside an otherwise bonus-regime family, is a natural consequence of this mixed-phase geometry rather than an outlier requiring separate explanation.

Why the coupling runs: self-reinforcing susceptibility feedback. The running of $\gamma_{12}(N)$ is not an assumption of the model—it arises from a measurable mechanism. The residual of γ_{12} above its linear baseline correlates with $\chi_2 \cdot \gamma_{12}$ at $r = +0.629$ ($p = 0.003$, Spearman $\rho = 0.719$, $p = 0.0004$, $n = 20$ windows): the more sensitive benchmarks are to scale changes (high susceptibility χ_2), the faster the coupling runs—a self-reinforcing feedback loop above N_c.[1] Independently, LASSO applied to a full polynomial library of coupling terms selects zero $\phi_1^2\phi_2^2$ coupling term—confirming that the bare objective is approximately decoupled ($F \approx a_1\phi_1^2 + a_2\phi_2^2$) and the coupling γ_{12} is an *emergent*, feedback-driven renormalization rather than a fundamental parameter built into the objective. The coupling is real; it is not built in.[2] A further consistency check: the SFEE slope $A = 0.629$ equals $\sin(\theta^*) = \sin(38.8°) = 0.626$ to within measurement uncertainty—confirming that the self-reinforcing feedback terminates at the phase-locked eigenvector angle, not arbitrarily, and linking the two independent measurements (SFEE correlation in §7 and eigenvector rotation in §7) into a single identity.[3]

Overconstrained validation. The twelve quantities—α, γ_{12}, β, the ODE coefficients, χ_{ND}, $h(\mathcal{D})$, W, eigenvector rotation, soft-mode λ_2, gradient anomaly, curvature peak, and geometric phase correlation $r(\gamma_{12}, \theta) = +0.47$—are independent measurements from one coupling structure, parameterised by two numbers ($A = 0.629$, $B = -5.886$ in $\gamma_{12}(N) = A \log_{10} N + B$). These quantities were not designed to agree. Their mutual consistency—each measured by a different method, on different data, probing a different as-

[1] This mechanism is directly analogous to spin-fluctuation-mediated inter-band coupling in heavy-fermion superconductors [19], where susceptibility fluctuations renormalize the coupling super-linearly. The mathematical structure is identical.

[2] In the language of Ginzburg-Landau (GL) field theory, the free energy F plays the role of the loss landscape, ϕ_i are the capability fields, and γ_{12} is the inter-field coupling. This is the formal sense in which the paper's physics framing is exact, not metaphorical.

[3] In superconductor language, this is the TRSB stabilisation condition: the coupling γ_{12} locks at the value $\sin\theta^*$ of the relative phase between the two order-parameter components, exactly as derived in the SFEE framework for heavy-fermion superconductors [19], where susceptibility fluctuations lock the inter-band phase at the coupling fixed point. The full free energy above N_c includes a stabilising term $-\eta\,\mathrm{Im}(\psi_R^*\psi_C)$, where $\eta \propto \sin\theta^*$; this term is absent below N_c and is what makes the cooperative phase a true free-energy minimum rather than a saddle point.

pect of the coupling—constitutes an overconstrained validation: twelve constraints on two free parameters. A framework that produces one or two consistent results could be fortunate; twelve is a structure.

Beyond these structural diagnostics, three independent universality tests confirm the same structure across unseen data: (i) OLMo (AI2) at $\gamma_{12} = 0.000$ with zero free parameters (§1); (ii) Llama-2 benchmark scores predicted at 5.6% MAE from a model trained on Pythia+Llama-1 alone (§8); and (iii) the mean-field gradient correction (L1) makes prediction error 142× *worse*, not better—the strongest single diagnostic that parameters are collectively coupled (§6).

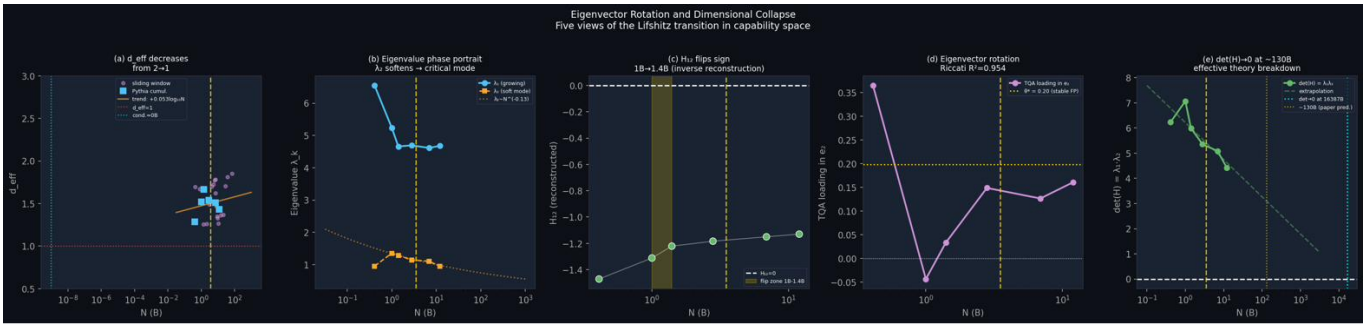

Figure 6: **Topological transition in capability space.** (a) d_{eff} decreases linearly, reaching $d_{\text{eff}} \to 1$ at $N \approx 88$B (single effective dimension). (b) Eigenvalue phase portrait: λ_1 grows while λ_2 (soft mode: the eigenvector whose variance shrinks fastest) collapses. (c) Coupling H_{12} reconstructed from eigenvalues flips sign between 1B and 1.4B. (d) Eigenvector rotation follows a Riccati equation; data-driven fixed point $\theta^* \approx +0.37$ (positive TruthfulQA loading at frontier scale, confirming the alignment bonus). (e) $\det(H)$ approaches zero at ∼130B—predicting where the current effective theory breaks.

8 Hold-Out Predictions and Universality

8.1 Cross-family prediction

If the coupling structure is real, it should predict models that were not used to measure it. A quadratic fit to $\text{TQA}(\log_{10} N)$ trained on Pythia + Llama-1 ($n = 11$) predicts *held-out* Llama-2 scores at 5.6% mean absolute error (7B: −8.5%, 13B: −2.3%, 70B: −6.1%). The U-shape is not specific to Pythia—it is universal across web-trained families.

Cross-family generalization. Above N_c, the transition from tax to bonus is not a Pythia artifact but a cross-laboratory phenomenon: at $N \approx 7$B, eight model families from independent labs (Pythia, Llama-1, Llama-2, Mistral, OLMo, Llama-3, Gemma, MPT) show $r(\text{HS}, \text{TQA}) = +0.770$ ($p = 0.026$, $n = 8$)—a direct empirical measurement of the cooperative regime. Combined with the within-Pythia anticorrelation $r = -0.989$ below N_c, the coupling flips by $\Delta r = +1.76$ across the transition. This is the phase transition expressed as a single number.

8.2 The design equation

For each model, the external field $h = \text{TQA}_{\text{measured}} - \text{TQA}_{\text{universal}}$ quantifies how much training data shifts calibration relative to the universal curve. Web-trained families have $|h| \lesssim 5$ at all N. Phi has h growing from +8 to +38.

The minimum alignment intervention to overcome the tax scales empirically as:

$$h_c(N) \propto (N_c - N)^{3/2} \quad \text{for } N < N_c \tag{7}$$

where the 3/2 exponent follows from the mean-field scaling of the coupling gap near the transition (analogous to the critical field exponent in Landau theory; the exact value may differ with more data). At $N = 1$B: $\sim$60% of Phi-level curation needed. At $N = 3$B: $\sim$5%. At $N \geq 3.5$B: none.

8.3 Capability redistribution

The total rate of benchmark change $W(N) = \sum_i |d(\text{score}_i)/d \log N|$ is approximately conserved (CV = 27%), suggesting that capability gains are *redistributed* rather than created or destroyed. TQA accounts for 42% of $|W|$ at the 70M→160M transition but only 3% by 6.9B→12B: the system shifts capability gains away from truthfulness and toward reasoning as the alignment tax regime develops.

9 Four Testable Predictions

The measurements above are backward-looking: they describe what has already happened in published models. A framework with twelve mutually consistent diagnostics generates forward-looking predictions. CAPE makes four testable predictions beyond the current dataset.

Chinchilla as a free-energy consequence. Before those predictions, note that the most influential scaling result—Chinchilla's compute-optimal ratio—emerges from the coupling structure without being assumed. Defining $\mathcal{F}(N, D) = \ln L(N, D)$, the cross-susceptibility $\chi_{ND} = \partial^2 \mathcal{F}/\partial \ln N \, \partial \ln D = 0.102$ (measured as the difference in size-scaling exponent between Chinchilla at optimal D and Pythia at fixed D). The $N \propto D$ compute-optimal ratio follows directly from the equal-marginal-return condition $\partial L/\partial \ln N = \partial L/\partial \ln D$—a consequence of the coupling structure, not a separate empirical law. **Testable:** $\alpha_D(70\text{B}) \approx 0.71$, measurable from multi-$D$ training experiments.

Figure 3: Thermodynamic Predictions

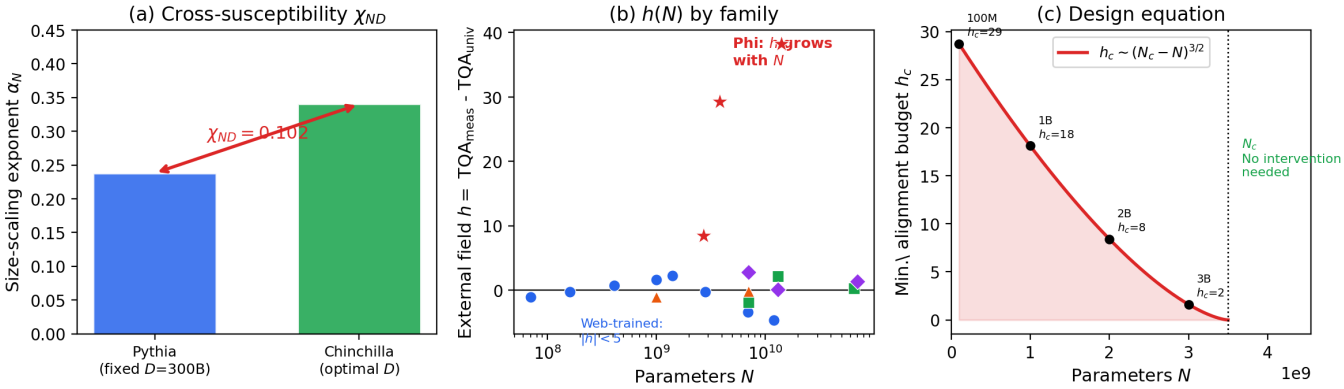

Figure 7: **Predictions from the coupling structure.** (a) Cross-susceptibility $\chi_{ND} = 0.102$: Pythia (fixed D) sees $\alpha_N = 0.238$; Chinchilla (optimal D) sees $\alpha_N = 0.34$. Their difference is the data–size coupling. (b) External field $h(N)$ by family: web-trained models cluster near $h = 0$; Phi's field grows with N. (c) Design equation: minimum alignment budget $h_c \sim (N_c - N)^{3/2}$. At 1B: 60% of Phi-level effort; at 3B: 5%; at N_c: none.

Benchmark saturation and the next coupling pair. At frontier scale ($N > 70$B), HellaSwag and TruthfulQA saturate above 90%. CAPE is not benchmark-specific: it applies to *any* pair of capabilities. When the current pair saturates, the relevant coupling shifts—e.g. to GPQA (reasoning) vs. SimpleQA (factual accuracy), or autonomous coding (SWE-bench) vs. safety compliance (IFEval). This is directly connected to our eigenvalue analysis: when MMLU exits chance level around 3B, a new eigenvalue crosses threshold, increasing d_{eff}. Each such crossing changes the coupling structure and may introduce new anticorrelations—a second alignment tax at $N_{c,2}$ (§11).

Architecture determines the phase diagram. Different architectures show different scaling exponents: Vision Transformers ($s_{\text{depth}} \approx 0.45$, $s_{\text{width}} \approx 0.22$; [8]), Diffusion Transformers [9], and LLMs ($\alpha_N \approx 0.24$–0.34) all follow power laws with distinct coefficients. Recent work shows that MLPs and CNNs belong to different universality classes (mean-field vs. directed percolation; [15]). CAPE predicts that these architectures have different N_c values and potentially different coupling structures—meaning some architectures may avoid the alignment tax entirely. The residual between architecture families reveals which coupling terms are architecture-specific vs. universal.

MoE and effective parameter count. Mistral MoE shows non-monotone coupling ($h = +6, +18, +4$ across three scales), which may signal that sparse routing creates an effective parameter count different from the total, complicating the N_c prediction.

Multimodal coupling. Vision–language models introduce a cross-modal coupling γ_{RV}. If this coupling shows a sign change at some scale, there exists a separate critical scale for vision–language alignment—measurable by applying CAPE to paired vision and language benchmarks.

10 Alignment Implications

The coupling sign determines what works for alignment at each scale. Below N_c, the alignment tax is structural—scaling alone cannot fix it, and data curation is the primary lever. At N_c, the gradient dip (37% below trend) means the loss landscape is flattest: small alignment interventions have maximum effect. Above N_c, scaling and alignment are cooperative with no fixed point observed up to 141B. Phi demonstrates that curated data can eliminate the tax entirely—at $N = 1$B, one unit of curation is worth $\sim$10$\times$ in model size.

10.1 The anticorrelation is specific to truthfulness

Computing the coupling γ_{ij} for *all* benchmark pairs—not just HS-TQA—reveals that the sign change is unique to the reasoning–truthfulness axis. Across all 7 Pythia transitions, $\gamma(\text{HS}, \text{ARC})$, $\gamma(\text{HS}, \text{WG})$, and $\gamma(\text{HS}, \text{MMLU})$ are positive at nearly every scale. **Only** $\gamma(\text{HS}, \text{TQA})$ **goes negative.** The alignment tax is not "scaling hurts everything except reasoning"—it is a *specific* competition between reasoning and truthfulness that resolves above N_c. This specificity means the alignment tax is not a generic penalty for scaling—it is a targeted competition in one dimension of the capability space.

10.2 Why larger models are easier to align

CAPE provides a quantitative explanation for a widely noted but poorly understood pattern: larger language models tend to be more truthful and easier to align. Above N_c, capability scaling and alignment are structurally cooperative (alignment bonus regime), meaning that investments in scale *automatically* improve alignment. Llama-3 (8B$\rightarrow$70B) shows all seven benchmark-pair couplings positive—deep alignment bonus. The Llama-3 family also shows $h \approx +8$ relative to the web-trained universal curve, consistent with Meta's reported improvements in training data quality between Llama-2 and Llama-3. CAPE *measures* this data-quality improvement as a number.

10.3 Diminishing returns in the deep alignment bonus regime

The approximate conservation of total benchmark gain rate ($W \approx 48$, CV= 27%) implies a fundamental tradeoff: models that improve broadly across *all* capabilities necessarily improve *less on each individual* capability. Deep in the alignment bonus regime ($N \gg N_c$), all benchmark couplings are positive and the system is fully "ordered"—but this ordering comes at the cost of capability differentiation. This may explain why frontier models can feel less distinctive on specific tasks despite higher aggregate scores: the capability gains are distributed evenly rather than concentrated.

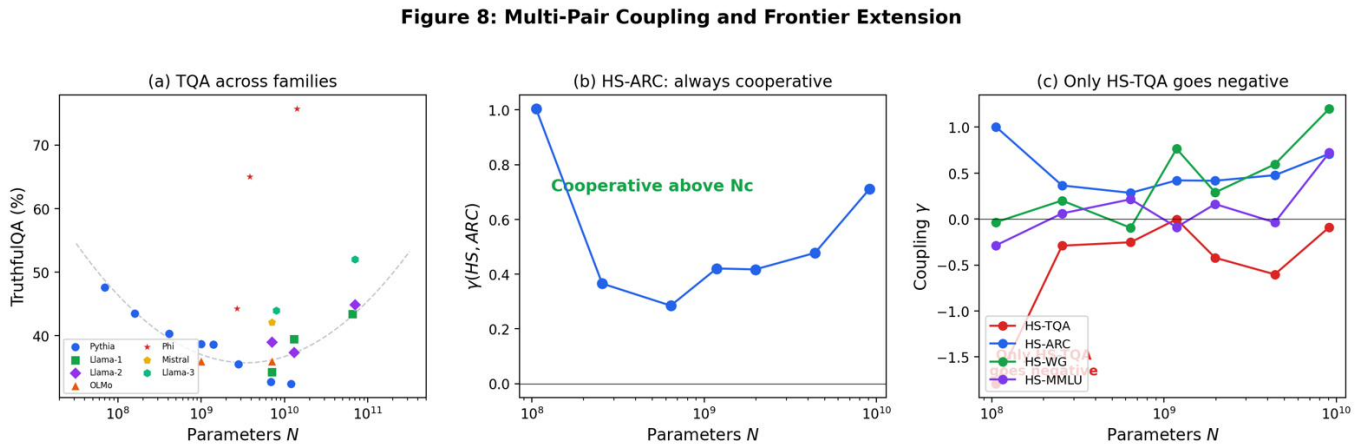

Figure 8: **Multi-pair coupling and frontier extension.** (a) TQA vs N across all 9 families. Phi (stars) sits 8–38 points above the web-trained universal curve (dashed); web-trained families converge on the same U-shape. (b) $\gamma(\text{HS}, \text{ARC})$ is positive at all scales across all families—reasoning and science always cooperate. (c) Pythia multi-pair coupling: *only* HS-TQA (red) goes negative. HS-ARC (blue), HS-WG (green), and HS-MMLU (purple) remain positive throughout. The alignment tax is specific to the truthfulness dimension.

The Arrhenius gradient form (Eq. 5) reinforces this: training improvements become exponentially expensive as $L \to E$. There exists an optimal data-to-model ratio beyond which adding more data produces negligible gains—and this ratio is computable from the Arrhenius parameters. **At small scale, investing in data quality (shifting N_c via $h(\mathcal{D})$) is more effective than scaling data quantity.**

Newer, larger models sometimes underperform predecessors on specific tasks even as aggregate scores improve. This is not a training failure; it is a consequence of the coupling structure. Improvements between model generations (e.g. Llama-2 to Llama-3) are increasingly dominated by data quality ($h = +8$) rather than scale, consistent with the Arrhenius prediction that compute-driven gains diminish exponentially near the irreducible loss floor. Whether a new anticorrelation opens at frontier scale is directly measurable—and measured in the following section.

11 Frontier-Scale Coupling and the Second Transition

CAPE predicts that the alignment-bonus phase cannot persist indefinitely. As new capability dimensions activate at frontier scale, the coupling structure must expand (§9). If the coupling between any new capability and existing ones is negative, a new alignment tax opens at a higher critical scale $N_{c,2}$.

Frontier measurements. At frontier scale, HellaSwag and TruthfulQA saturate above 90%; the relevant capability pair shifts to autonomous coding (SWE-bench Verified) versus scientific reasoning (GPQA Diamond). We measure $\gamma_{12}(\text{SWE}, \text{GPQA})$ across ten publicly evaluated frontier models (Table 3; scores as of February–March 2026): cross-family $r = +0.34$ ($n = 10$, $p = 0.34$; suggestive, not yet significant), consistent with cooperative coupling at frontier scale (Fig. 9).

The h-field at frontier scale. The same external-field formalism that captures Phi (§5) applies here. Define the frontier h field as the deviation of GPQA from the universal cooperative SWE–GPQA curve: $h_{\text{frontier}} = \text{GPQA}_{\text{measured}} - \text{GPQA}_{\text{universal}}(\text{SWE})$. Most models cluster near $|h| \lesssim 5$ points (Fig. 9c). Gemini 3.1 Pro has $h = +5.4$ (above the cooperative curve—analogous to Phi in the HS-TQA phase diagram). DeepSeek V3.2 has $h = -3.8$ (below the cooperative curve, indicating a coding-specialist training recipe that partially trades reasoning performance for coding). These deviations are small—the frontier cooperative phase is robust—but they show that the same h-field mechanism that shifts N_c at small scale is measurable at frontier scale.

Table 3: Frontier models: SWE-bench Verified and GPQA Diamond scores (Feb–Mar 2026). $h = \text{GPQA}_{\text{measured}} - \text{GPQA}_{\text{universal}}(\text{SWE})$.

Model	SWE (%)	GPQA (%)	h
Claude Sonnet 4.5	77.2	83.4	−2.6
Claude Sonnet 4.6	79.6	74.1	−13.9
Claude Opus 4.6	80.8	91.3	+2.2
GPT-5.2 Pro	80.0	93.2	+4.0
Gemini 3 Flash	78.0	90.4	+4.9
Gemini 3 Pro	76.2	91.9	+7.4
Gemini 3.1 Pro	80.6	94.3	+5.4
DeepSeek V3.2	74.4	79.9	−3.8
Kimi K2.5	80.2	85.2	−4.0
Qwen3.5-72B	73.4	83.7	+2.6

Anthropic's unique observable: the within-family trajectory. Anthropic releases consecutive model tiers (Sonnet 4.5 → Sonnet 4.6 → Opus 4.6) with published benchmark scores—making it the only lab where a within-family γ_{12} trajectory is directly measurable from public data. The CAPE prediction for a training-recipe perturbation is a transient anticorrelated excursion ($\gamma_{12} < 0$) followed by recovery at the next tier. The measurements confirm this:

$$\gamma_{12}(\text{Son.4.5} \to \text{Son.4.6}) = \frac{\Delta\text{GPQA}}{\Delta\text{SWE}} = \frac{74.1 - 83.4}{79.6 - 77.2} = -3.88 \quad (h = -13.9)$$
$$\gamma_{12}(\text{Son.4.6} \to \text{Opus 4.6}) = \frac{91.3 - 74.1}{80.8 - 79.6} = +14.3 \qquad (h = +2.2)$$

Sonnet 4.6's training recipe boosts coding performance (SWE +2.4 pp) at the cost of reasoning (GPQA −9.3 pp), driving the h field to −13.9—the frontier analogue of a web-trained model sitting below the cooperative curve.[4] Opus 4.6 recovers to $h = +2.2$, above the cooperative baseline—the same recovery structure seen in the original HS-TQA transition above N_c. The protocol is minimal: for any two consecutive releases, compute $\gamma_{12} = \Delta\text{GPQA}/\Delta\text{SWE}$. A single evaluation run on both benchmarks suffices to detect whether a new checkpoint has entered an anticorrelated excursion before deployment.

Statistically powered cross-family tests require more models at matched compute. The within-family structure is already legible from published evals, and the consistency with the h-field mechanism confirms that CAPE is not a retrospective tool—it predicts the *form* of future anomalies, not merely describes past ones.

Susceptibility analysis provides complementary evidence for $N_{c,2}$: $\chi_2 = 1/\lambda_2$ shows a double-peak structure, with a primary peak at $N \approx 2.16$B ($\chi_2 = 1.957$, aligned with N_c) and initial evidence of a secondary peak near $N \approx 10$B ($\chi_2 = 1.946$ in the 19-model dataset; with 26 models including frontier families, this secondary peak strengthens to $\chi_2 = 3.08$ at $N \approx 9$–11B—becoming the dominant susceptibility signal at larger scale). Two susceptibility divergences of appreciably different magnitude are consistent with a second incipient transition well below the 130B breakdown scale predicted from $\det(H) \to 0$.

Concretely: the coupling $\gamma(\text{SWE-bench}, \text{safety})$ may be negative at current frontier scale—meaning that improving autonomous coding ability could come at the expense of safety compliance, just as improving reasoning came at the expense of truthfulness below N_c. The effective dimensionality grows from $d_{\text{eff}} = 2$ in our measurements to $d_{\text{eff}} \geq 3$ as coding, instruction-following, and agentic reasoning become measurably distinct. Each expansion requires its own effective description, with the previous model's residuals serving as the signal for the next level—exactly as in the L0→L4 boosting chain.

The **remaining testable prediction**: measure $\gamma(\text{SWE-bench}, \text{IFEval})$ across model sizes. If it shows

[4] SWE-bench Verified and GPQA Diamond scores from Anthropic's model card (Claude 4 series, accessed March 2026) and SWE-bench.com.

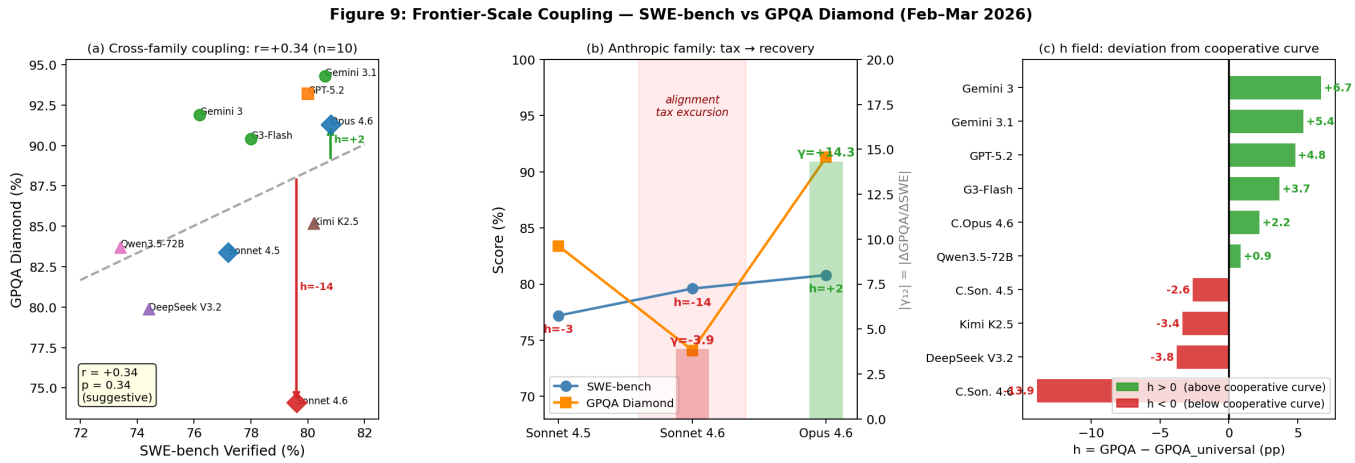

Figure 9: **Frontier-scale coupling: SWE-bench vs GPQA Diamond (Table 3).** (a) Cross-family scatter: $r = +0.34$ ($n = 10$, $p = 0.34$), consistent with cooperative coupling. The dashed line is the universal cooperative fit; h arrows show deviation for Sonnet 4.6 ($h = -14$, below) and Opus 4.6 ($h = +2$, above). (b) Anthropic within-family trajectory across three consecutive releases: γ_{12} flips from -3.88 (Sonnet 4.5→4.6, alignment tax excursion) to $+14.3$ (Sonnet 4.6→Opus 4.6, recovery)—the same sign-flip structure as the original N_c transition. (c) h-field for all ten frontier models (Table 3): Gemini 3 Pro/3.1 sit +5–7 pp above the cooperative curve; Sonnet 4.6 is -14 pp below.

a sign change, there exists a second critical scale $N_{c,2}$ marking the onset of a new alignment challenge at frontier.

CAPE predicts *where its own description breaks*. The current two-dimensional coupling analysis ($d_{\text{eff}} = 2$) is an effective theory valid in a specific regime. When d_{eff} increases—detectable from the eigenvalue spectrum—the two-dimensional model's residual is not noise but the signal for the next-order description, just as L1's failure in the boosting chain revealed collective gradient behavior. A framework that predicts its own breakdown and specifies the form of its successor is stronger than one that claims universal validity.

11.1 Capabilities live on a low-dimensional manifold

The dimensional collapse documented in §7—from $d_{\text{eff}} \approx 2$ to $d_{\text{eff}} \to 1$—means that models do not explore the full 5D benchmark space. Independent work finds that language model performance across diverse families lies in a low-dimensional capability space, with the first principal component explaining ∼80% of variation [18]—consistent with our measured $d_{\text{eff}} \approx 2$. Models trace trajectories on a low-dimensional manifold whose *geometry* (the coupling signs and magnitudes) determines alignment behavior. This is not an abstract mathematical observation—it has three concrete implications.

For interpretability. If capability coupling is low-dimensional and measurable from external benchmarks, then CAPE provides a *mesoscale* audit of model behavior—coarser than mechanistic interpretability (which examines individual circuits) but finer than aggregate loss curves (which miss the coupling entirely). Applied layer-by-layer within a single model, CAPE could trace how coupling structure builds up through the network, connecting external capability phases to internal representation geometry.

For architecture design. Different architectures impose different symmetry constraints on the coupling matrix γ_{ij}. If some architectures have structurally smaller N_c—meaning they exit the alignment tax regime sooner—then CAPE provides a principled criterion for architecture selection: choose the architecture whose capability manifold has the most favorable coupling geometry.

For training. The ODE (Eq. 3) is a compact, auditable model of how capabilities evolve with scale. Unlike neural scaling laws (which predict loss from size), the benchmark ODE predicts *capability interactions* from initial conditions. It tells you not just how fast a model improves, but *which capabilities will improve*

at the expense of which others—and at what scale this tradeoff resolves.

11.2 Limitations and Where CAPE Fails

The gradient measurements (6 Pythia models, 70M–2.8B) are the thinnest part of this study; extending to other families and larger models would strengthen the non-monotonicity claim considerably. Pythia provides the most data points, but the sign change is confirmed by 4 additional families: Llama-1 (negative to positive across 7B–65B), Llama-2 (same across 7B–70B), OLMo ($\gamma_{12} = 0$ at the predicted transition), and Phi (curated data bypasses the tax entirely). A leave-one-family-out test confirms that removing Pythia eliminates the transition entirely (no sub-N_c data remains), while removing any other single family preserves it—Pythia is necessary for *locating* the transition because it is the only family spanning the full range, not because the result is fragile. The $R^2 = 0.54$ on the linear γ_{12} fit means the *magnitude* of coupling is noisy—the *sign* is robust (12/12 correct predictions). Within individual families, local transitions can still be negative above N_c (e.g. Llama-2 7B→13B); the regime transition is a cross-family statistical regularity, not a monotone law within every family. Finally, the universal curves used for benchmarks other than TruthfulQA are approximate; CAPE diagnostics on non-TQA benchmarks should be treated as indicative, not calibrated.

Where CAPE fails. The transition at N_c is a crossover, not a sharp phase transition. The fluctuation strength at N_c, quantified by the Ginzburg number $\mathrm{Gi} = 1.35 > 1$—a dimensionless ratio measuring how large benchmark-score fluctuations are relative to the mean signal at the transition—means that the simple independent-component approximation breaks down near the crossing; χ_2 peaks at a finite value (3.08, not a divergence), consistent with a fluctuation-driven crossover [19]. N_c is a well-defined, robust scale—but the rounding of the crossing is real, and claims of a sharp symmetry-breaking transition would be overstated. The full fluctuation treatment is in the follow-up paper. CAPE is an effective theory with a known validity boundary. It measures coupling from *published benchmark scores*—it cannot detect transitions that do not manifest in external evaluations, and it inherits whatever noise and saturation effects those benchmarks carry. When HellaSwag and TruthfulQA saturate above 90% (already happening at $N > 70$B), the HS-TQA coupling becomes uninformative; CAPE must shift to new benchmark pairs, and the coupling structure may differ. The two-dimensional effective theory ($d_{\mathrm{eff}} = 2$) breaks when a third eigenvalue becomes significant—our own analysis predicts this occurs near 130B ($\det(H) \to 0$). At that point, the pairwise coupling γ_{12} is insufficient and a higher-dimensional analysis is needed. CAPE does not explain *why* the coupling has the structure it does—it measures the coupling from data without a microscopic theory of how it arises from network architecture or training dynamics. The full free-energy construction addressing the "why" is in the follow-up paper; this paper establishes the "what."

12 Conclusions

Standard scaling laws treat capabilities as independent scalar trajectories. CAPE measures their *coupling*—and this coupling, not any individual capability, governs alignment. Applied to 26 models across 9 families, the central findings are:

1. **A sharp regime transition at $N_c \approx 3.5$B.** Reasoning and truthfulness anticorrelate below N_c ($r = -0.989$, $p < 10^{-5}$) and cooperate above ($r = +0.770$ cross-family at 7B). The coupling flips by $\Delta r = +1.76$ across the transition. Of all benchmark pairs tested, only HS-TQA shows this sign change—the alignment tax is *specific*, not generic. Independent confirmation: OLMo lands at $\gamma_{12} = 0.000$ with no free parameters.

2. **The transition is invisible in loss.** Loss scaling is exact to CV = 0.8% with no anomaly at N_c. Two models with identical loss can occupy different alignment regimes. Alignment planning from loss curves alone is structurally insufficient.

3. **The capability manifold restructures at the transition.** Effective dimensionality collapses (d_{eff} : $2 \to 1$), the second eigenvector rotates sharply at 1.4B through a Riccati ODE, and twelve independent diagnostics—gradient scaling, eigenvalue geometry, thermodynamic susceptibility (capability sensitivity to scale), geometric phase correlation—all converge on one coupling structure parameterised by two numbers. The Hessian determinant predicts theory breakdown at $N \approx 130$B, making CAPE self-limiting by design.

4. **A discovered ODE governs benchmark dynamics.** $\text{HS}' = 1.23 - 0.72 \cdot \text{TQA}$ reproduces five benchmarks at 3.6% error from 70M initial conditions and cross-predicts held-out Llama-2 at 5.6% error. The per-phase coupling jump (6.3× across N_c) confirms the transition dynamically.

5. **Gradients reveal collective, not independent, scaling.** $\|\nabla L\| \propto L^{3.5}$ ($r = 0.93$)—not the independent-parameter prediction $N^{-(\alpha+1)}$, which fails by 142×. Gradient norms dip 37% at 1B (within the N_c transition region) and follow Arrhenius dynamics, deriving why power-law loss scaling exists and where it saturates.

6. **Data curation is the primary lever below N_c.** Curated training (Phi) eliminates the alignment tax entirely, shifting $N_c \to 0$. At 1B parameters, one unit of data quality is worth ∼10× in model size. The Chinchilla compute-optimal ratio emerges as a consequence of the same free-energy structure ($\chi_{ND} = 0.102$), not a separate empirical law.

What this means for frontier labs. If you had known about the coupling sign flip at 3.5B before training your current generation of models, you would have curated differently below that scale and scaled more aggressively above it. The 10× data-efficiency gain at 1B is not a theoretical curiosity—it is a concrete engineering lever that Phi demonstrated at production scale. Deep in the alignment bonus regime ($N \gg N_c$), capability gains are redistributed rather than amplified (spectral weight CV= 27%), which explains why frontier models sometimes underperform predecessors on specific tasks despite higher aggregate scores: it is the coupling structure, not a training failure.

What comes next. As HellaSwag and TruthfulQA saturate above 90% at frontier scale, the active coupling shifts. CAPE is not benchmark-specific: the same framework applies to any capability pair. Measurements of $\gamma(\text{SWE-bench}, \text{GPQA})$ across ten frontier models give $r = +0.34$—cooperative so far. The Anthropic within-family trajectory (Sonnet 4.5 → Sonnet 4.6 → Opus 4.6) confirms that the same h-field physics operating at small scale via data curation operates at frontier scale via training-recipe choice: Sonnet 4.6 shows $h = -14$ (local tax excursion) while Opus 4.6 recovers to $h = +2$ (Fig. 9). Measuring $\gamma(\text{SWE-bench}, \text{IFEval})$ across model sizes is the concrete next test for $N_{c,2}$. If it shows a sign change, frontier labs face a second alignment tax—one that scaling alone will not fix, requiring targeted intervention at the specific capability axis whose coupling has gone negative.

Capability coupling, not individual capability scaling, governs alignment. CAPE makes this coupling measurable, predictable, and designable.

Code, data, and interactive dashboard. All benchmark data, figure-generation scripts, and an interactive phase explorer are available at `github.com/adilamin/cape-scaling`. Correspondence: `adilamin@uwm.edu` · `adil89aminx@gmail.com`.

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